Spikes Dynamics in Fixed-Frequency Radio Emission

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Overview

1. Background and Purpose
2. Data
3. Methods
4. Discussion
5. References

Background and Purpose

- On **October 28, 2003**, an **X17.2 solar flare** from **NOAA AR 0486** generated one of the most powerful radio emissions recorded at the cycle 23.
- These emissions showed rapid intensity fluctuations known as **spikes**.
- Traditional linear analysis captures frequency and power but misses hidden dynamic behavior.
- This study uses **nonlinear time-series analysis** to identify whether these variations are **deterministic or random**.
- The goal is to describe the **complexity and possible chaotic nature** of these solar radio signals.

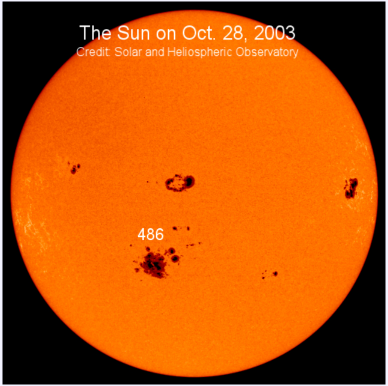


Fig. 1. Active Region 0486 during the Oct 28, 2003 flare (SOHO/EIT).

Data

- **Observatory:** INAF - Trieste Astronomical Observatory.
- **Frequencies:** 237 MHz, 327 MHz, 408 MHz (fixed-frequency).
- **Sampling:** 1000 data points/s; 4000-point stationary intervals selected.
- **Preprocessing:** background removal and rescaling.

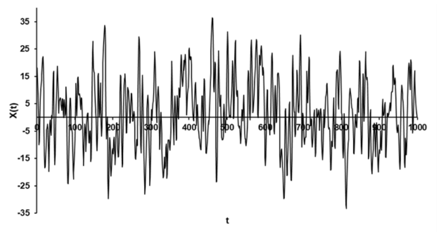


Fig. 2. Background-removed signal segment at 237 MHz.

Phase-Space Reconstruction

- Phase space via time-delay embedding (Takens, 1981):

  $$\mathbf{x}(t) = \left[x(t),\, x(t+\tau),\, \ldots,\, x(t+(m-1)\tau)\right]$$

- τ: first minimum of average mutual information.
- m: embedding dimension determined using the false nearest neighbors.

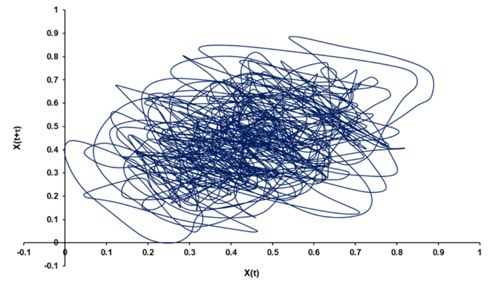


Fig. 3. Phase space attractor (237 MHz).

Nonlinear Time Series Analysis

Parameters & Physical Meaning

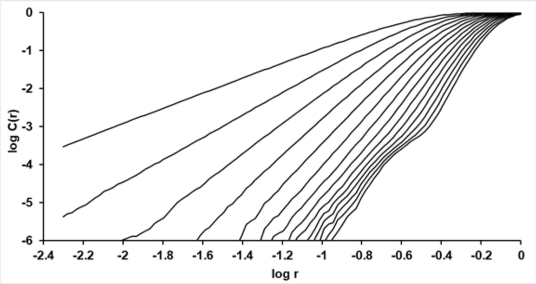
- **Lyapunov exponent** (λ): Sensitivity to initial conditions.
- **Hurst exponent** (H): Indicates long-range dependence.
- **Deterministic factor** (Δ): Predictability via recurrence plot determinism.
- **Correlation dimension** (D_c): Global fractal complexity.
- **Pointwise dimension** (D_p): Local scaling variability.

Nonlinear metrics across frequencies.

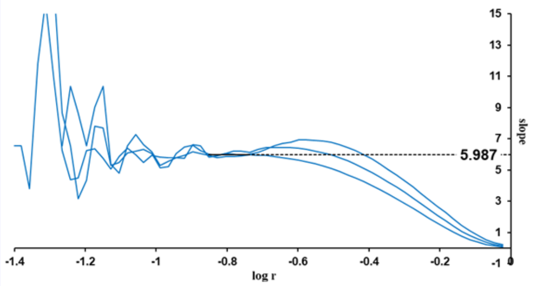
Freq	λ	Δ	H	D_c	D_p
237 MHz	0.449	0.708	0.267	6.017 ± 0.114	6.793 ± 0.524
327 MHz	0.279	0.678	0.257	6.097 ± 0.129	6.964 ± 0.558
408 MHz	0.560	0.623	0.263	6.055 ± 0.100	6.733 ± 0.477

Fractal Analysis Visuals: Correlation Dimension

- Slope of $\log C(r)$ vs. $\log r$ in the scaling region; the derivative $d \log C(r)/d \log r$ shows a **plateau** $\approx D_c$.



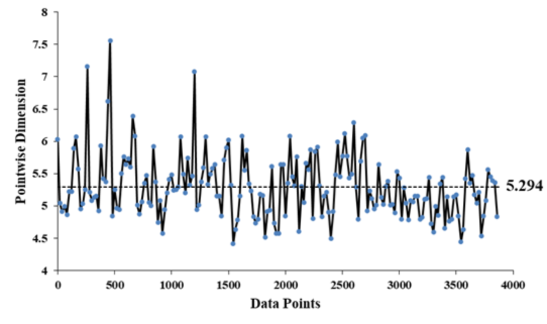
Correlation Integral $\log C(r)$ vs. $\log r$. Linear slope in the scaling window $\to D_c$.



Derivative $d \log C(r)/d \log r$. Plateau value $\approx D_c$; highlights the scaling range.

Fractal Analysis Visuals: Pointwise Dimension

- Unlike the correlation dimension D_c, which gives a global estimate, D_p describes how the complexity changes between different regions of the system.



Distribution of Pointwise Dimension D_p. Spread in D_p values shows local variability in scaling behavior.

Interpretation:

- Confirms spikes exhibit complex, non-uniform dynamics.
- High D_p variability complements D_c results.
- Indicates deterministic yet irregular behavior across the time series.

Results & Discussion

- The nonlinear analysis confirms that spikes exhibit **deterministic chaotic behavior**.
- The positive Lyapunov exponents and high deterministic factors indicate the presence of **sensitive dependence on initial conditions** and structured dynamics.
- Correlation dimension ($D_c \approx 6.0$) and pointwise dimension ($D_p \approx 6.7$–6.9) reveal a **high-dimensional attractor**, suggesting complex but organized variability.
- Hurst exponents ($H \approx 0.26$) show weak persistence, supporting the interpretation of intermittent and irregular temporal evolution.
- Together, these metrics confirm that the observed fluctuations are not random noise but reflect an underlying **chaotic deterministic process**.

Interpretation: Spikes behave as high-dimensional chaotic systems, where complexity arises from deterministic, but unpredictable, dynamics.

References & Acknowledgements

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